



Professional Summary:

- **Full Stack Developer | Generative AI & Applied ML professional with 15+ years of overall IT experience (Angular, ReactJs, Python, AWS, Nodejs, Java Spring Boot, Dotnet Core) and 3.5+ years of hands-on expertise in Generative AI, LLMs, RAG, Fine-Tuning, NLP, AWS Bedrock, Machine Learning, and Deep Learning, delivering enterprise-ready AI solutions.**
- **Proven experience building LLM-powered applications for document intelligence, semantic search, chatbots, and automated knowledge retrieval, with a strong focus on accuracy, relevance, and hallucination reduction.**
- **Strong hands-on expertise in Prompt Engineering, model evaluation, and optimization, leveraging AWS Bedrock, RAG architectures, and vector databases (Pinecone, FAISS, Chroma) to build scalable, context-aware AI systems.**
- **Hands-on experience working with LLM and embedding models including GPT-4o Mini, Gemini, Claude, LLaMA, OpenAI Embeddings, Gemini Embeddings, and Hugging Face Sentence Transformers to support semantic search and RAG pipelines.**
- **Knowledge in developing Agentic AI solutions using LangChain, LangGraph, and CrewAI, with working knowledge of Model Context Protocol (MCP) for context management and agent orchestration.**
- **Extensive domain experience delivering enterprise AI solutions across Airlines, Payments, Healthcare with strong understanding of compliance, data sensitivity, and domain-specific workflows.**

Technical Knowledge:

Scripting Languages: Python, SQL, PL/SQL, Shell Scripting

Frontend Frameworks: Angular 17+, React.js 18+

Backend & Full Stack: Node.js, Python, .Net Core, Java Spring Boot

Generative AI: Generative AI, Large Language Models (LLMs), Small Language Models (SMLs), Prompt Engineering, Prompt Optimization

Frameworks & Tools: LangChain, CrewAI, LangGraph, LangSmith, LCEL

LLM Models & APIs: AWS Bedrock, Llama 3 (Meta), Claude (Anthropic), OpenAI API, Gemini API, Groq

RAG: RAG Pipelines, Semantic Search, Chunking, Index Strategies

Vector Databases: Pinecone, FAISS, ChromaDB

ML / AI Libraries: Hugging Face Transformers

Cloud Services: AWS Lambda, ECS, Docker, Kubernetes, AWS CDK, AWS Bedrock

Databases: SQL, Oracle PL/SQL, Oracle 9i/10g

Monitoring & Observability Tools: Datadog, Dynatrace, Kibana

Career Experience

Project Lead, May 2015 to Current – (Mphasis)

Project Description:

1. Delta Airlines is modernizing its **TGM web applications** and broader **airline management systems** by migrating to AWS with the support of **AWS Migration Competency Partners**. The initiative aims to reduce operational costs, improve agility, and strengthen security while enabling advanced AI-driven capabilities.

To transform airline operations and workforce management, the project incorporates **Generative AI, Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG)**.

These technologies enhance decision-making across crew scheduling, passenger services, and talent management by analyzing **skills, experience, and role requirements** in real time.

Technologies: Angular, Nodejs, AWS Lambda Function, AWS CDK Pipeline, Python, Generative AI, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Semantic Matching, Prompt Engineering, FAISS, Pinecone

Roles & Responsibility:

- Designed and implemented **Retrieval-Augmented Generation (RAG) pipelines** using vector databases such as FAISS and Pinecone to enable **semantic passenger-to-service matching**, improving personalization and reducing hallucinations in recommendations.
 - Developed and optimized **LLM-based applications** (GPT, LLaMA, Claude, Gemini) to analyze passenger profiles, travel history, and preferences for **tailored service offerings and seat/crew assignments**.
 - Applied **prompt engineering and optimization techniques** to generate **fitment scores** for passenger needs (e.g., meal preferences, seat upgrades, loyalty benefits), along with explainable recommendations.
 - **Built semantic search capabilities to match passenger queries with relevant flight information, policies, and personalized offers in real time.**
 - Collaborated with **airline operations and customer service teams** to align AI-driven solutions with enterprise workflows, compliance requirements, and passenger experience strategies.
 - Leveraged **cloud-native services (AWS Bedrock, Lambda, ECS, Kubernetes)** to deploy scalable, secure, and resilient AI solutions for airline management systems.
2. United Airlines, in collaboration with **SAUA**, initiated the migration and modernization of **28 applications surrounding UCB** to the **AWS Cloud**. The project aimed to reduce infrastructure costs, improve agility, enhance security, and accelerate digital transformation across airline management systems. To maximize business value, the modernization effort incorporated **Generative AI and Retrieval-Augmented Generation (RAG)** to analyze **passenger information, airline data, and alliance workflows**. These AI-driven solutions enabled intelligent passenger matching, personalized service recommendations, and operational efficiency across the airline ecosystem.

Technologies: Angular, ReactJs, Nodejs, Harness Pipeline, Python, Dotnet Core, Java Spring Boot, AWS, LangChain, Retrieval-Augmented Generation (RAG), Pinecone, Semantic Search, Prompt Engineering, Git

Roles & Responsibility:

- Designed and implemented a Generative AI-based Passenger solution using AWS Bedrock and LangChain to support natural language interaction with multi-format airline documents, including passenger policies, loyalty programs, and service guidelines.
- Built and maintained the end-to-end **RAG lifecycle**, including passenger data ingestion, semantic chunking, embedding generation, vector indexing, and retrieval using FAISS/Pinecone, enabling accurate and grounded responses to passenger queries.
- Applied **prompt engineering and optimization techniques** to improve answer relevance, consistency, and contextual accuracy while minimizing hallucinations in passenger service recommendations.
- Developed **semantic search capabilities** to match passenger profiles with flight information, seat upgrades, loyalty benefits, and personalized offers.
- Collaborated with **airline operations, customer service, and compliance teams** to align the solution with enterprise workflows, aviation security standards, and alliance governance requirements (Star Alliance, United Airlines).

3. Bio-Rad DBG Group's **SeqSense NGS Data Analysis** is a cloud-native platform that removes the need for costly compute infrastructure, enabling large-scale RNA/DNA processing without servers. The pipeline, built on the **STAR ultrafast RNA-seq aligner** and optimized for the **SEQuoia Complete Stranded RNA Library Prep Kit**, ensures speed and precision in RNA-Seq analysis.

With **Generative AI integration**, SeqSense delivers automated insights, adaptive pipeline optimization, and natural language interaction. GenAI enriches results with biological context and scales seamlessly in the cloud, transforming SeqSense into a **powerful AI-driven discovery platform** for genomics.

Technologies: AWS Bedrock, LangChain, FAISS/Pinecone, Hugging Face, Python, ReactJs, Spark, STAR RNA-seq aligner, SEQuoia RNA Library Prep Kit.

Roles & Responsibility:

- Designed and implemented a **Generative AI-enabled bioinformatics solution** using AWS Bedrock and LangChain to support natural language interaction with multi-format genomic data, including RNA/DNA sequencing outputs, library prep protocols, and analysis guidelines.
- Built and maintained the end-to-end **RAG lifecycle** for genomic datasets, including data ingestion, semantic chunking, embedding generation, vector indexing, and retrieval using FAISS/Pinecone, enabling accurate and grounded responses to researcher queries.
- Applied **prompt engineering and optimization techniques** to improve contextual accuracy, reduce hallucinations, and enhance the relevance of AI-generated interpretations of RNA-Seq results.
- Developed **semantic search capabilities** to match sequencing profiles with experimental conditions, differential expression results, pathway enrichment, and downstream biological insights.
- Collaborated with bioinformatics scientists, cloud engineering teams, and compliance stakeholders to align the solution with enterprise workflows, genomic data security standards, and regulatory requirements.

Previous Experience

- Senior Software Engineer, Dec 2013 to Feb 2015 (SuhaSoft Pvt Ltd)
- System Dev Analyst, July 2013 to Dec 2013 (Dell International Global Services Ltd)

- Senior Software Programmer, May 2012 to May 2013 (Ventech Solutions)
- Senior Software Engineer, Dec 2010 to May 2012 (Source Hov Ltd)

Certifications

- Certified in AWS Solutions Architect – Associate
- Certified in AWS DevOps Engineer Professional

Education:

- MCA – 2004 to 2007 – University of Madras
- BBA– 2001 to 2004 – University of Madras